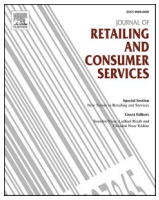
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From awareness to preference: The dual role of brand and eco-friendliness in shaping green choices

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ABSTRACT

Given the growing popularity of online shopping, promoting environmental awareness and encouraging eco-friendly choices on these platforms are essential for advancing sustainability. This study explores the specific influence of presenting environmental educational video messages right before product selection, focusing on how they enhance visual attention to eco-friendly information and influence price sensitivity, brand impact, and preferences for green products among Generation Z consumers. The research is grounded in a theoretical framework integrating the Elaboration Likelihood Model (ELM), Priming Effect, and Theory of Planned Behavior (TPB). It comprises two studies that combine eye-tracking technology with data collected by questionnaire. To analyze eye-tracking data, this study employs a novel analytical method by integrating Dynamic Grey Relational Analysis (DGRA) with T-tests, providing an advanced approach to evaluate behavioral patterns, ranking variable importance, and addressing non-linearity in consumer attention metrics beyond traditional significance thresholds. Based on the priming effect, the findings indicate that the timing of these messages significantly enhances attention to eco-friendly attributes, reduces price sensitivity, and guides decision-making toward greener choices by emphasizing the importance of sustainability. Additionally, the study highlights the dual role of brand familiarity, showing that under heightened environmental awareness, it strengthens preferences for eco-friendly products while diminishing its influence on non-eco-friendly ones. These insights provide practical implications for marketers and policymakers, emphasizing the strategic use of environmental video messages in digital retail to drive sustainable consumption and reinforcing the importance for brands to align with environmental values to sustain consumer engagement and remain competitive in increasingly sustainability-driven markets.

1. Introduction

In recent years, concerns about environmental degradation and sustainability have attracted significant global attention, leading to the emergence of “green consumption" as a potential solution. This trend is driven by an increasing awareness among individuals regarding the environmental impacts of their consumption behaviors, thereby motivating them to make more eco-friendly decisions. Green consumption involves mindful selection, purchasing, and using products and services that minimize environmental harm throughout their life cycle (Shehawy and Khan, 2024; Wei et al., 2024). A key requirement for promoting green consumption is increasing environmental awareness, which involves understanding environmental issues, recognizing its importance to human life, and acknowledging the need for proactive efforts to protect the environment (Alhamad et al., 2023).

Despite the growth in consumer knowledge regarding green products, awareness remains relatively low in emerging markets, presenting significant challenges in promoting sustainable choices. One major barrier is the higher price of green products compared to conventional alternatives, which complicates the decision-making process for consumers. When there is a lack of sustainability information, consumers often prioritize price over environmental benefits, which decreases the likelihood of purchasing eco-friendly products. However, as more information on sustainability becomes available, enhancing environmental awareness influences consumers’ sensitivity to product information, making them more likely to consider both price and environmentally significant attributes, thereby allowing them to balance these factors in their purchasing decisions (Cook et al., 2023; Meise et al., 2014; Pagiaslis and Krontalis, 2014; Shehawy, 2023). This shows that positive environmental attitudes play an important role in

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influencing green consumption behaviors, beyond just economic factors (Alhamad et al., 2023). Furthermore, consumers are more inclined to adopt pro-environmental behaviors when such actions are consistent with their personal values and interests (Eid et al., 2021).

Additionally, brand recognition plays an important role in shaping consumer trust in a company, further encouraging sustainable consumption (García-Salirrosas et al., 2024). Consumers with environmental knowledge are more likely to purchase a brand when they perceive it as eco-friendly (Belloque et al., 2023). According to Punyatoya (2014), improving consumer perceptions of a brand's environmental sustainability coupled with elevated environmental awareness, can increase their willingness to purchase eco-friendly products (Punyatoya, 2014). On the other hand, research suggests that increased environmental awareness can reduce the impact of brand familiarity for non-eco-friendly brands while strengthening it for eco-friendly brands, ultimately leading consumer decisions to more sustainable choices (Ewe and Tjiptono, 2023; Rustam et al., 2020).

The rise of e-commerce, an internet-based commerce, has transformed the retail industry. It enables customers, retailers, and distributors to engage in online transactions for purchasing, selling, and financial exchanges. The rapid growth of e-commerce has led to the widespread availability of online shopping platforms, giving consumers more opportunities to discover eco-friendly products (Jauhar et al., 2024; Y. Zhang et al., 2024). In this context, consumer engagement becomes crucial for promoting sustainable behaviors, particularly in online shopping environments where interactions are often brief (Chen et al., 2020). Furthermore, advancements in internet technology provide innovative ways to disseminate environmental information, enhance awareness, and promote sustainable actions (Yang et al., 2021).

Moreover, previous studies have shown that attitudes, subjective norms, perceived behavioral control, green thinking, online education, and online marketing positively influence the intention to purchase green products on digital platforms (Nekmahmud et al., 2022). Consumers with environmental concerns often display strong positive attitudes toward eco-friendly products or services, particularly on online platforms. These attitudes are significantly shaped by the emotional component of the value-attitude system, which involves emotional responses to environmental issues (Park et al., 2021). Additionally, exposure to environmental messages, whether through traditional media or online platforms, has been found to significantly enhance individuals' sustainable attitudes and positively affect their behaviors toward the environment (Meng et al., 2023). Research indicates that video-based messages are more effective than text-based ones in improving perceptions of environmental issues, making them a valuable tool for raising environmental awareness (Goldberg et al., 2019). According to the priming effect, exposure to environmental messages unconsciously influences subsequent decisions, fostering more environmentally sustainable behaviors. Additionally, the timing of such messages plays a crucial role in capturing consumer attention and shaping behavior (Tate et al., 2014; S. Zhang et al., 2024; Zsidó, 2024).

While most research has focused on consumer perception and attitudes using traditional methodologies, integrating those methods with neuroscience can reveal more profound insights into the hidden aspects of the decision-making process (Crespo-Pereira et al., 2022). Traditional research methods, such as questionnaires and interviews, often fail to capture implicit and unconscious consumer behaviors. In contrast, neuromarketing and consumer neuroscience offer a more accurate depiction of consumer behavior by revealing these underlying processes (Bhardwaj et al., 2023; Xu et al., 2023). As a result, combining neuromarketing with traditional research methods has become increasingly common, blending explicit measurement tools with innovative approaches to uncover consumers' genuine emotions and preferences (Casado-Aranda et al., 2023).

The retail sector has experienced a profound transformation, creating various opportunities for online retailers. In this digital marketplace, consumer neuroscience can play an important role in identifying the factors that attract consumers and influence their purchasing behaviors (Yadete and Kant, 2023). Eye-tracking technology is widely used in neuromarketing to gain an understanding of consumer behavior by analyzing areas of interest and predicting high-engagement zones. Examining eye fixation patterns provides valuable insights into cognitive processes, as visual attention plays a key role in how individuals process information. Research has shown a positive correlation between visual attention, comprehension performance, and decision-making outcomes, making eye-tracking a powerful tool for enhancing consumer engagement through targeted marketing strategies (Li et al., 2025; Liu et al., 2022; Song and Jiang, 2018).

The study was conducted over six weeks in a laboratory setting at the College of Economics and Management, Nanjing University of Aeronautics and Astronautics, with participants from different nationalities within Generation Z. This research investigates the impact of environmental educational video messages presented right before the product selection phase on price sensitivity and consumer purchasing behavior toward green products in an online shopping context. It addresses the gap regarding the timing of presenting green stimuli and its effect on green purchasing behavior, a topic that has been underexplored in existing studies within this field.

While previous research has mainly focused on the positive role of brand influence in encouraging green consumption, the two-way effects of brand influence in different situations and at different levels of environmental awareness still require further exploration. This study also aims to fill this gap by examining how brand familiarity (whether it is eco-friendly or not), interacts with environmental awareness in shaping green product preferences, under conditions of heightened environmental awareness.

The study integrated eye-tracking technology and the questionnaire as data collection methods. Eye-tracking technology was selected to capture implicit consumer responses, particularly related to visual attention to eco-friendly attributes (Kim and Kim, 2024; Zamani et al., 2016), while the questionnaire collected self-reported data on consumer preferences and attitudes. This combination of methods allowed for a more comprehensive analysis of consumer behavior, capturing both conscious and subconscious responses (Casado-Aranda et al., 2023). Potential biases were reduced through the stratified random sampling approach, which ensured that gender and income levels were balanced across the two experimental groups (Espigares-Jurado et al., 2020). Moreover, participants were randomly assigned to either the video-exposure group or the control group, further minimizing any selection bias in the study. Additionally, the research uses DGRA and T-test to enhance methodological robustness and account for non-parametric variations in consumer responses.

By extending the theoretical foundations of the ELM and Priming Effect, which focus on the impact of exposure to stimuli and its timing on attitudes (S. Zhang et al., 2024; Zsidó, 2024), along with the TPB, which examines the impact of attitudes on choices (Eid et al., 2021), this study provides new insights into the effects of environmental message timing and the role of brand familiarity on sustainable consumption choices.

2. Theoretical background and hypothesis development

This study explores the impact of environmental educational video messages presented right before the product selection phase on online shopping platforms. It aims to understand how these messages influence price sensitivity and consumer engagement with green products and explore their effect on purchasing behavior. Specifically, the research investigates the role of brand influence in different situations in consumer green behavior across varying levels of environmental awareness.

To explore these relationships, the study addresses the following research questions.

1. How does the presentation of environmental educational video messages right before the product selection phase impact consumer

visual attention to various product attributes and preferences for green products?
2. How does exposure to environmental educational messages influence consumer price sensitivity in the context of green purchase intention?
3. How does the interaction between brand familiarity and environmental awareness affect consumer decision-making in relation to green purchase intention?
4. What is the impact of brand influence on consumer price sensitivity in the context of green purchase intention?

By addressing these questions, the study aims to provide valuable insights into the psychological mechanisms that drive sustainable consumption choices in online shopping environments. The findings will offer practical implications for online green retailers and policymakers aiming to enhance consumer engagement with green products through targeted educational interventions and digital marketing strategies.

2.1. Environmental challenges and the shift towards green consumption

Technological progress and economic development have significantly improved living standards. However, this advancement has also led to increased environmental challenges. This dual effect has resulted in widespread concern among nations about addressing the challenges arising from the intersection of technological advancement, economic growth, and environmental impact (Niu et al., 2022). One key strategy for mitigating these challenges is encouraging individuals to adopt environmentally conscious consumption behaviors. There is a complex relationship between the preferences of the younger generation, conscious consumption patterns, societal influences, corporate actions, and the need for effective environmental governance. Within this interconnected framework, the importance of individual awareness, mindful consumption choices, corporate accountability, and addressing global environmental challenges becomes increasingly urgent. Green consumption is a significant element of environmentally friendly behaviors, highlighting the importance of conscious decision-making in minimizing ecological impact (Daoud et al., 2024). Greater awareness of environmental issues enhances understanding of the importance of protecting the environment for human health and preserving the planet. Promoting environmental awareness requires a thorough understanding of these challenges and serves as an effective strategy to encourage pro-environmental behaviors and foster green consumption (Darvishmotevali and Altinay, 2022; Maduku, 2024). Given these challenges, raising environmental awareness is critical for fostering a shift towards green consumption, enabling individuals to make informed choices that minimize their environmental impact.

2.2. Improving environmental awareness in online platforms through video messages

Environmental awareness involves recognizing various environmental issues and understanding the necessary measures to adopt practices that support environmental preservation. It encompasses a wide range of perspectives, including opinions, concerns, and viewpoints on environmental challenges, as well as ideas and strategies for addressing these challenges (Dabbous et al., 2023). Previous studies have emphasized that environmental awareness is a fundamental factor in promoting environmental sustainability and protection. It plays a crucial role in shaping individuals' attitudes toward the environment and enhancing their level of consciousness regarding environmental issues, which, in turn, significantly influences their purchasing intentions (Li et al., 2020; Rustam et al., 2020). Shehawy (2023) reported that environmental knowledge moderates the relationship between consumers' intentions to purchase green products and their actual buying behavior. This suggests that a consumer's awareness of environmental issues can influence how they attend to product information, thereby helping to explain the observed connection (Shehawy, 2023).

Several studies have shown that internet usage affects individuals' perceptions of social fairness, health, and overall quality of life. In the environmental domain, the literature explores the impact of internet usage on fostering pro-environmental behavior, satisfaction with government environmental efforts, and perceptions of environmental risks or quality (Zhang and Gong, 2023). Sun et al. (2020) found that social media advertising significantly influences product knowledge, perceived consumer effectiveness, attitudes and intentions related to purchasing green products (Sun and Wang, 2020). Liu et al. (2021) discovered that internet use not only directly impacts pro-environmental behaviors but also indirectly affects them through mediators such as environmental knowledge, perceived pollution threats, and government protection efforts (Liu et al., 2021). In a report on March 27, 2023, the China Annual Conference on Environmental Communication emphasized the importance of using social media as a key strategy for promoting environmental protection. They suggested that increasing the circulation of environmentally focused content such as news, scientific presentations, documentaries, and educational materials on online platforms could significantly enhance public environmental awareness (Meng et al., 2023).

Furthermore, unlike textual content, videos can present complex information more clearly, making them powerful tools for promoting products and influencing purchasing behavior. Effectively using video content can foster positive attitudes toward environmental sustainability (Kurisu et al., 2019). According to Goldberg et al. (2019), videos that present the scientific consensus on global warming significantly enhance public perception compared to written transcripts containing the same information (Goldberg et al., 2019). Utilizing the capabilities of online platforms is an effective solution for increasing environmental awareness, as it can shape pro-environmental attitudes and encourage sustainable consumption behaviors. In light of this, the present study examines the impact of incorporating educational environmental video messages into online shopping platforms.

2.3. The impact of price sensitivity on green purchasing behavior

Price sensitivity, which refers to how a product's price affects consumers' purchasing decisions, is a key factor influencing green purchasing behavior (Lavuri, 2022; C. Li et al., 2023). The willingness to pay more for eco-friendly products is still a topic of debate, as some consumers believe that producers and governments should offer sustainable products without charging high prices (Oesman, 2021). In a study by Srivastava and Gupta (2023), the authors found that price sensitivity negatively affected green purchase intentions, especially in relation to perceived behavioral control. Their findings suggested that price sensitivity could weaken the influence of other factors on the intention to purchase green products (Srivastava and Gupta, 2023).

On the other hand, Barbu and colleagues (2022) reported that individuals are more willing to pay a higher price for green products if they perceive the products' characteristics, designs, and functions as beneficial to themselves or the environment (Barbu et al., 2022). Similarly, Minbashrazgah et al. (2017) reported that environmentally aware consumers are more responsive to sustainable information, which motivates them to purchase higher-priced green products (Minbashrazgah et al., 2017). Ansu-Mensah (2021) suggested that educating consumers about the benefits of investing in green products is crucial, as it could increase their willingness to pay more for these items (Ansu-Mensah, 2021). Based on this information, one of the aims of this study is to examine the impact of increasing environmental awareness on the price sensitivity of green products.

2.4. Neuromarketing, eye tracking and visual attention

Neuromarketing involves assessing consumers' psychological and sensory responses to marketing communication. By exploring the human

brain, neuromarketing provides deeper insights into how individuals intuitively respond to marketing activities, offering novel information that traditional methods cannot provide. Recently, neuromarketing has become increasingly popular in both academic and commercial contexts, significantly enhancing our understanding of consumer behavior and promoting sustainable consumption. In terms of sustainable consumption, neuromarketing has played a crucial role in helping businesses to market sustainable products effectively. This highlights the importance of understanding consumer psychology and behavior in ways that extend beyond traditional marketing approaches (Singh et al., 2023).

Eye-tracking is a neuroscience tool that measures visual attention by tracking where individuals focus their gaze on presented stimuli. This process relies on the visual mechanism of selective attention, where the brain focuses on specific objects or areas while simultaneously suppressing attention to non-selected areas. This approach is connected to theories regarding the determinants of attention, helping researchers gather distinct information about stimuli that are relevant to marketing effects (Kim and Kim, 2024; S. Li et al., 2023). For instance, in a study conducted by Alonso et al. (2018), the researchers assessed the visual appeal of main web page images to digital consumers (Alonso and Sánchez, 2018). Similarly, Zamani et al. (2016) investigated the influence of participants' visual attention on product preferences by integrating cognitive science related to decision-making with eye-tracking techniques (Zamani et al., 2016).

This study employs a combination of eye-tracking technology and questionnaires to obtain more accurate insights. By focusing on Generation Z, a demographic known for its advanced digital proficiency and greater environmental awareness (Manchanda et al., 2023), this research aims to expand existing theories on consumer engagement in green marketing and enhance participation in green consumption. These traits make Generation Z particularly relevant for studying green consumption behaviors, as this cohort is more likely to integrate environmental values into their purchasing decisions and show responsiveness to digital marketing strategies, especially those focused on sustainability.

2.5. Elaboration Likelihood Model and priming effect in green behavior

The Elaboration Likelihood Model (ELM) suggests that when individuals process a message and subsequently change their attitudes, the elaboration level and motivation influence whether they engage in central or peripheral route processing. Central route processing focuses on the content of the message through a cognitive approach. In contrast, peripheral route processing relies on superficial cues and engages in an emotional approach when motivation or ability to process the message is low (Luo et al., 2024; Mosler and Martens, 2008; Petty et al., 1997). According to Zhang et al. (2024), peripheral cues require less cognitive effort and mainly depend on emotional triggers. Affective appeals embedded within a message evoke emotions that shape attitudes and behaviors (S. Zhang et al., 2024). In line with this, the ELM also posits that these routes of persuasion can shape individuals' attitudes toward green advertising and influence their intentions to engage in environmentally friendly consumption (Marroquín-Ciendúa et al., 2020). In the context of this study, video-based environmental messages serve as a tool to elicit emotional responses, thereby engaging peripheral processing, while simultaneously promoting cognitive engagement through the central route to increase consumer environmental awareness and influence decision-making.

The Priming Effect refers to the unconscious influence that initial exposure to a stimulus has on responses to subsequent stimuli. It often involves emotional arousal, which can affect visual attention. According to emotional contagion theory, emotions can be transferred and modified through messages, ultimately shaping the recipient's attitudes and behaviors. Studies indicate that the timing of exposure to these stimuli significantly impacts the level of visual attention (Lindblom, 2023; S. Zhang et al., 2024; Zsidó, 2024). Tate et al. (2014) found that the Priming Effect facilitates an automatic positive evaluation of stimuli relevant to a person's goals, leading to more environmentally sustainable decisions (Tate et al., 2014). Additionally, Bimonte et al. (2020) found that the Priming Effect increases the prominence of pro-environmental attitudes. The type of framing used also affects the willingness to pay a premium for environmentally friendly products and the amount of that premium (Bimonte et al., 2020).

In this study, environmental educational videos serve as priming stimuli, timed just before the product selection phase to evoke emotions that engage peripheral processing, enhance environmental awareness through central route processing, and influence consumer decisions. The theories discussed suggest that exposure to environmental content can activate subconscious attitudes and shape consumer preferences, particularly in relation to green consumption. Based on this understanding, the following hypotheses have been developed in response to research questions 1 and 2.

H1. Exposure to environmental educational messages right before the product selection phase triggers emotional arousal, which enhances visual attention to eco-friendly information and eco-labels.

H2. Increased visual attention to eco-friendly information significantly enhances consumers' preference for sustainable products over non-eco-friendly products, regardless of brand influence.

H3. Exposure to environmental educational messages right before the product selection reduces price sensitivity by moderating visual attention.

2.6. Theory of Planned Behavior in consumer green consciousness and brand familiarity

According to the Theory of Planned Behavior (TPB), attitudes, subjective norms, and perceived behavioral control are crucial factors influencing an individual's intention to perform a specific behavior. In recent decades, this theory has been recognized as a reliable and effective model for predicting and analyzing individual environmental behaviors (Al Mamun et al., 2018; García-Salirrosas et al., 2024). In the context of green consumption, consumers are more likely to engage in pro-environmental behaviors when this behavior aligns with their personal interests. For example, individuals with heightened environmental awareness often have more positive attitudes toward green products, making them more likely to purchase sustainable items. They also experience increased normative pressure to engage in eco-friendly behaviors and perceive a greater sense of control over their choices. These elements significantly increase consumers' intention to purchase eco-friendly products (Choi and Johnson, 2019; Eid et al., 2021; García-Salirrosas et al., 2024).

Previous studies have suggested that heightened environmental awareness enhances consumers' environmentally conscious behavior and green consciousness (Li et al., 2020; Rustam et al., 2020). Consumer green consciousness refers to the degree of consumer attentiveness to environmental issues and their willingness to take action to address these concerns. Based on Ewe et al. (2023), empirical findings indicate that green consciousness may reduce the impact of brand familiarity when consumers are faced with a non-eco-friendly brand. Conversely, it may increase the influence of brand familiarity when consumers are faced with an eco-friendly brand (Ewe and Tjiptono, 2023). Based on the TPB and supporting evidence, the following hypothesis has been formulated to examine how increased environmental awareness moderates the effect of brand influence on consumer preference under different situations, particularly by shaping attitudes, strengthening subjective norms, and enhancing perceived behavioral control in green purchasing decisions, thereby addressing research question 3.

H4. Environmental awareness moderates the effect of brand influence on consumer preference.

H4a. When the product is eco-friendly, environmental awareness strengthens the positive impact of a well-known brand on consumer preference.

H4b. When the product is non-eco-friendly, environmental awareness weakens the impact of a well-known brand on consumer preference.

2.7. Brand awareness and its influence on green consumption

Brand awareness plays a significant role in consumer decision-making and market performance. Aaker (1996)defines it as the extent to which consumers can recognize or recall a brand in various situations. This awareness is essential for shaping the initial consideration set of brands that consumers evaluate when making purchase decisions (Abidin and Panjaitan, 2024). Research by Abdullah and colleagues (2023) indicates that consumers often prioritize factors such as quality, strong brand associations, brand image, brand familiarity, and unique attributes over price. This suggests that these elements can influence purchase intentions, extending beyond price sensitivity (Abdullah et al., 2023).

Additionally, substantial evidence shows that familiar brands have a significant communication advantage over unfamiliar ones (Delgado-Ballester et al., 2012). Belloque et al. (2023) found that brand awareness significantly influences consumer behavior, leading to more sustainable choices. Their study revealed that participants showed a stronger preference for sustainable brands after being informed about their eco-friendly attributes (Belloque et al., 2023). According to Yang (2017), consumers with environmental concerns are positively influenced by both green products and brand awareness in their purchasing decisions (Yang, 2017). Furthermore, Oanh (2024) reports that when consumers have confidence in a familiar brand's commitment to sustainability, they are more likely to be motivated to purchase that brand's products (Oanh, 2024). Based on these findings, the following hypothesis has been formulated in response to research questions 3 and 4.

H5. Under conditions of heightened environmental awareness, consumers show stronger engagement with eco-friendly products from familiar brands compared to non-branded alternatives.

H6. Brand influence reduces consumers' price sensitivity, as evidenced by decreased visual attention to the price factor when encountering a familiar brand.

The Study Framework is illustrated in Fig. 1.

3. Method and experiment details

A six-week laboratory study was conducted using the SMI iView RED 4 eye-tracker and a questionnaire to investigate the hypotheses. The selected products for the study included both eco-friendly and non-eco-friendly items, as well as branded (familiar) and unbranded options.

According to Najafabadiha et al. (2025), a prevention-based educational video message was developed to raise environmental awareness, as this framing has shown a greater impact (Najafabadiha et al., 2025). This 2-min and 14-s video presented information based on data from the World Health Organization (WHO) and scientific research, emphasizing the benefits of green products in preventing and reducing pollution and environmental degradation. It featured imagery of environmental pollution and wildlife affected by waste and human activities, with key messages strategically integrated into the background to underscore the urgency of sustainable consumption.

The experiment comprised two separate studies, each involving three products. The visual content and informational materials presented to participants were carefully selected from Amazon to ensure that the experiment was conducted in an authentic context relevant to the featured products.

3.1. The procedures of the experiments were as follows

The participant grouping process started with data collection through a questionnaire conducted over one week. Participants were informed about using an eye-tracking device for gathering data and provided their informed consent to participate in the experiment. After grouping, participants submitted their socio-demographic information as the first step of the study. In the next phase, one group of participants viewed a video message right before accessing the product pages, while the other group proceeded to the next stage without any exposure to the message.

In Study 1, data collection began with participants engaging with the study objects, which included sales pages for products A, B, and C on an

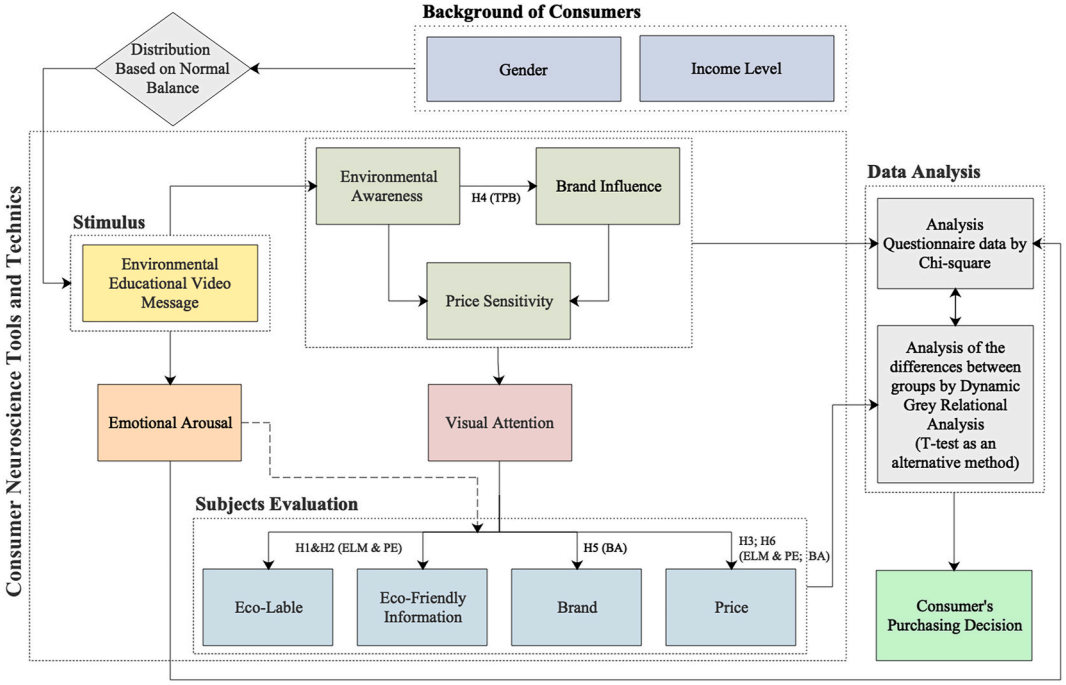


Fig. 1. Study framework.

online shopping platform. Participants could easily navigate these pages by pressing the spacebar to access and review the content. Afterward, they answered questions about their purchase intentions, various related variables, and their eco-friendly purchasing habits. Study 2 followed the same procedure but focused on products D, E, and F. All questions were displayed on a computer monitor using the SMI iView Experiment Center software, and participants submitted their responses via mouse clicks. The experimental procedure is illustrated in Fig. 2.

3.2. Study population

Participants from Generation Z were chosen for the study due to their heightened permeability to changes in consumer behavior, digital skills, greater environmental awareness, commitment to equality and justice, and ethical diversity, with many shared behavioral traits across cultures (Manchanda et al., 2023). The study involved 112 participants recruited through both online and offline announcements. Out of 112 participants, one hundred met the criteria and had normal or corrected-to-normal vision, with a nearly even distribution of gender, 51 % female and 49 % male.

Considering the subscriptions of Generation Z with participants from different nationalities (Manchanda et al., 2023), the research employed a stratified random sampling approach to ensure balance in gender and income levels, two influential factors in purchase intention (Migheli, 2021). The stratified sampling ensured that the groups were balanced in these variables, allowing for a more generalized understanding of the findings across different demographic groups. Initially, socio-demographic information was collected from the participants, and based on a balance of the two previously mentioned factors, they were randomly assigned to the experimental groups (G1 and G2) without any bias regarding their identity. This method was adapted from techniques previously used by Espigares-Jurado and colleagues (2020) to meet the requirements of this research (Espigares-Jurado et al., 2020). This sampling methodology was designed to maintain the randomness necessary for generalizability and statistical validity.

A priori power analysis was conducted using G*Power software (Faul et al., 2007) to determine the necessary sample size for the study. This analysis focused on a T-test involving two groups, assuming an effect size of $f = .5$, a significance level of .05, and a statistical power of .80. The results indicated that a total sample size of 100 participants would be sufficient. Therefore, including 100 participants in this study ensures adequate statistical power to detect significant differences, which aligns with established guidelines for sample sizes in eye-tracking research (Deng and Gao, 2023; Sharafi et al., 2020).

3.3. Data collection

Each participant completed the experiment individually in a quiet room, free from external distractions. The first group (G1) consisted of participants who did not watch a video, while the second group (G2) included participants exposed to the video message. In both studies, participants evaluated three different products that varied in eco-friendliness, brand, and price.

Study 1 involved unisex hoodies (Products A, B, and C):
- *Product A:* An eco-friendly, familiar brand (most expensive)
- *Product B:* Neither branded nor eco-friendly (lowest price)
- *Product C:* Eco-friendly but not branded (priced between Products A and B)

Study 2 involved razors (Products D, E, and F):
- *Product D:* Neither branded nor eco-friendly (lowest price)
- *Product E:* Eco-friendly but not branded (most expensive)
- *Product F:* Familiar brand but not eco-friendly (priced between Products D and E)

After completing the eye-tracking procedure, participants were asked to fill out the questionnaire, which included questions on purchase intentions and other relevant variables. Based on the approach described by Luan and colleagues (Luan et al., 2023), purchase intention was evaluated through the question, “How likely are you to purchase product A (D)/B(E)/C(F) in the future?" Socio-demographic information for all participants was collected through a questionnaire. Responses were measured on a 5-point Likert scale, where 1 indicated “not at all" and 5 represented “extremely."

3.4. Eye movement analyses

The raw data collected using the SMI iView Experiment Center version 3.6 software were transferred to the SMI iView BeGaze version 3.6 software for analysis. To assess attention distribution across stimuli, the product page was labeled into distinct Areas of Interest (AOIs), which included the Product Info, Price, Eco-Label, and Brand. The analysis focused on four primary metrics commonly used in eye-tracking studies: First Fixation Duration, Revisits, Fixation Count, and Fixation Time. Each metric was calculated for every AOI to derive the final results.

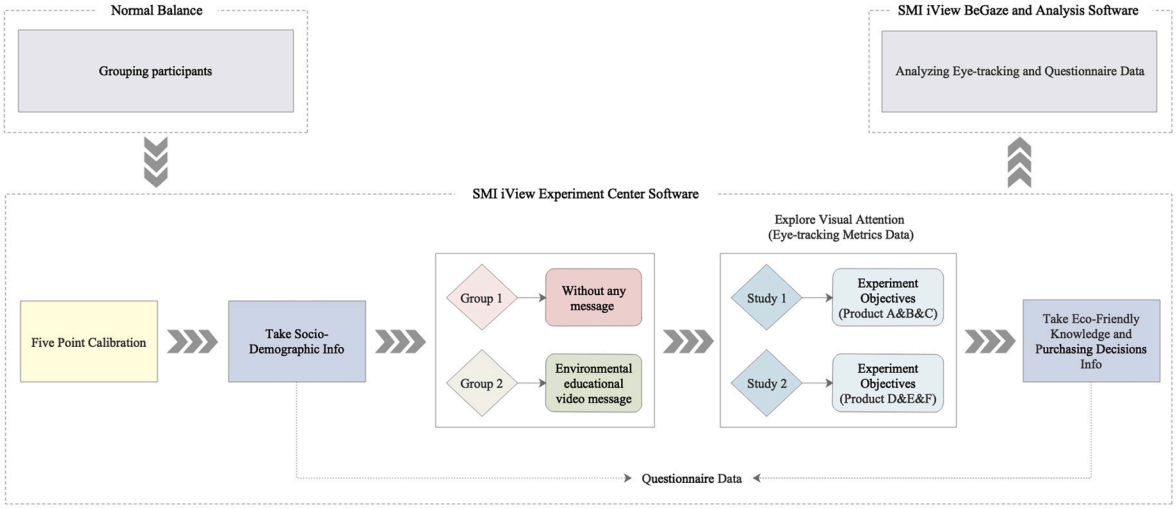


Fig. 2. The procedure of the experiment.

3.5. Statistical analysis

In this study, the combination of Dynamic Grey Relational Analysis (DGRA) techniques and the T-test (results are presented in Tables 1 and 2) was chosen due to the unique strengths of each method in providing a comprehensive analysis of consumer behavior. In DGRA, the use of the Grey Relational Variator (ψ(j)) and Dynamic Distinguishing Coefficient (ξ(j)) allows for the dynamic simulation of changes in relationships between criteria under different conditions (such as time or specific circumstances), which is not possible in traditional GRA. This feature helps the DGRA model create more complex and non-linear relationships between criteria, whereas traditional GRA is only capable of analyzing linear and fixed relationships. Moreover, by using dynamic coefficients, DGRA reduces errors arising from fluctuations or random changes in the data, especially in datasets with non-normal characteristics, small sample sizes, or temporal variations, which cannot be effectively modeled in traditional GRA. As a result, DGRA provides higher accuracy in analyses due to its flexibility and adaptability to complex and unstable data (Javanmardi et al., 2021; Liu, 2025; Tu et al., 2024). Recent statistical discussions have underscored the limitations of relying solely on p-values for inferential decision-making, particularly in behavioral research where nuanced patterns and practical significance often outweigh rigid statistical thresholds. Traditional significance testing, while valuable, can overlook meaningful relationships that may not meet arbitrary cutoffs (Imbens, 2021). Rafi et al. (2020) suggest that the main challenges stem from human psychology and social influences, where researchers often rely on traditional statistical frameworks built on mistaken reasoning and limited understanding. These issues underscore the need to develop more advanced statistical approaches, clearer interpretive standards, and terminology that are less susceptible to misunderstanding or misuse than those passed down through conventional practice (Rafi and Greenland, 2020). As emphasized by McShane et al. (2024), statistical significance should not be determined solely based on p-values. Instead, the interpretation of results should incorporate additional statistical metrics, effect sizes, and the broader research context. Rather than treating p-values as binary decision criteria, study results should be assessed using multiple statistical approaches, including point and interval estimates, and should be interpreted alongside complementary methods. Furthermore, conclusions should be drawn based on cumulative evidence, integrating multiple statistical indicators holistically rather than relying on a single threshold value (McShane et al., 2024).

In this study, while some T-test results did not fall below the conventional p-value threshold for statistical significance, the DGRA analysis effectively addressed these gaps, explicitly overcoming the limitations inherent in p-value reliance by ensuring practically significant relationships are identified even when traditional significance thresholds are not met. DGRA offers significant advantages, particularly in its ability to capture complex, non-linear relationships and handle datasets with non-normal distributions or small sample sizes, where parametric tests such as T-tests may be less reliable. Unlike standard effect size metrics, DGRA provides a dynamic ranking of variable importance, allowing for a deeper understanding of relative influences in consumer behavior. This approach ensures that findings with practical significance are not dismissed simply due to p-value constraints, reinforcing the robustness of results even in cases where traditional significance thresholds are not met (Javanmardi et al., 2021; Liu, 2025).

By employing DGRA in conjunction with T-tests, this study aligns with best practices in contemporary statistical analysis, ensuring a multi-faceted interpretation of data. This approach mitigates the risks of misinterpretation associated with sole reliance on p-values and emphasizes a more holistic and context-driven assessment of findings. Such methodological integration is crucial in behavioral studies, where consumer decision-making is influenced by subtle cognitive and emotional factors that may not always be captured through conventional significance testing alone.

In addition to ranking and simulating relationships between variables, DGRA has the ability to calculate the Grey Relational Distance. This feature allows the analyst to measure the extent of changes and sensitivity in the relationships between variables. In other words, DGRA can act as a sensitivity analysis tool, helping to identify the variables that have the most significant impact on the analysis results and providing a better understanding of the relationships that are more sensitive to small changes in the data. This feature is especially useful in situations where the data exhibits fluctuations or uncertainty, contributing to a more precise analysis of relationships and the importance of each variable (Javanmardi et al., 2021; Tu et al., 2024).

In this study, sensitivity analysis using DGRA is employed to examine how changes in data impact the stability of consumer outcomes through the analysis of visual attention data. This approach analyzes data collected from consumers' unconscious reactions to various factors, including environmental information, price, and brand cues. Furthermore, it helps to identify the conditions under which consumer preferences for green products either significantly change or remain stable, such as slight variations in attention to different elements. Given the complex and non-linear nature of consumer behavior, sensitivity analysis in this research illustrates how random fluctuations in the data can influence the intricate relationships between variables and consumer preferences. This capability to analyze uncertainty and evaluate stability is particularly crucial when dealing with small sample sizes and data characterized by mental variability and non-linearity.

The study approach involves calculating Grey Relational Grades (Γ, GRG) for each group across each criterion and comparing them. The GRG is generally computed as a weighted average of Grey Relational Coefficients (GRCs), providing a positive measure of the degree of association. A higher GRG indicates greater focus and attention from group participants in each factor.

If $X_0 = (x_0(1), x_0(2), \cdots, x_0(n))$ is an ideal (reference) data set and $X_k = (x_k(1), x_k(2), \cdots, x_k(n))$, $k = 1, 2, 3, \cdots, m$, alternative data sets of equal length to be compared with it, then the GRG assesses the correlation between matching points in the ideal and alternative datasets. This "grey correlation" is influenced by four factors, with three being constants, providing a structured approach to measuring the relationship between datasets within the grey system theory framework. These factors include:

- Absolute Difference between ideal and alternative data sets: $|x_0(j) - x_k(j)|$
- The largest of all Absolute Difference values: $\max_k \max_j |x_0(j) - x_k(j)|$
- The smallest of all Absolute Difference values: $\min_k \min_j |x_0(j) - x_k(j)|$
- The Distinguishing Coefficient: ξ

Given that three of the factors in the Grey Relational Coefficient (GRC) formula are constants, the value of GRC is mainly determined by the Absolute Difference. This makes the Absolute Difference the critical variable component in the formula, significantly influencing the GRC's outcome. The GRG is given by

$$\Gamma_{0k} = \gamma(X_0, X_k) = \frac{1}{n}\sum_{j=1}^{n} \gamma_{0k}(j) \tag{1}$$

whereas the GRC is given by

$$\gamma_{0k}(j) = \gamma(x_0(j), x_k(j)) = \frac{\min_k \min_j |x_0(j) - x_k(j)| + \xi \ \max_k \max_j |x_0(j) - x_k(j)|}{|x_0(j) - x_k(j)| + \xi \ \max_k \max_j |x_0(j) - x_k(j)|} \tag{2}$$

The most ambiguous aspect of traditional GRA model is ξ, whose definition, range, or method for determining its optimal value has never been conclusively established. This oversight could be due to scholars commonly assuming $\xi = 0.5$. This assumption has solidified into a tradition that is rarely questioned (Mahmoudi et al., 2020).

Table 1
DGRA and T-TEST comparison of AOIs among two groups in the study 1.

Product			Product A					Product B					Product C					Averages of Metrics	Averages of AOIs	
			DGRA	T-TEST				DGRA	T-TEST				DGRA	T-TEST				DGRA		
AOI	Metric	GRP	GRG	Mean	SD	T	p-v	GRG	Mean	SD	T	p-v	GRG	Mean	SD	T	p-v	GRG	GRP	GRG
Product Info	FFD	G1	.556	245.66	56.454	−2.281	.025	.557	222.96	55.561	2.186	.031	.511	210.12	55.124	−2.777	.007	.541	G1	.511
		G2	.609	271.56	57.08			.503	196.58	64.757			.605	248.34	80.182			.572		
	RV	G1	.494	2.82	1.699	−1.465	.146	.518	2.62	1.64	1.533	.129	.493	2.52	1.515	−1.696	.093	.502		
		G2	.519	3.36	1.977			.499	2.16	1.346			.525	3.04	1.551			.515		
	FC	G1	.484	5.34	2.429	−2.764	.007	.522	5.1	2.621	1.405	.163	.486	4.92	2.431	−2.802	.006	.497	G2	.532
		G2	.527	7.34	4.502			.496	4.44	2.042			.534	6.3	2.493			.519		
	FT	G1	.479	2316.4	801.893	−1.981	.051	.536	2154.9	931.39	1.988	.05	.502	2034.46	1002.149	−2.163	.033	.506		
		G2	.539	2709.16	1149.941			.477	1812.4	785.621			.552	2477.36	1044.954			.523		
Price	FFD	G1	.56	149.16	45.71	1.305	.195	.569	170.76	50.451	2.614	.01	.531	147.08	46.477	1.056	.294	.553	G1	.556
		G2	.525	136.9	48.169			.504	145.04	47.905			.493	138.02	39.01			.507		
	RV	G1	.563	4	1.512	3.943	<.001	.55	4.18	2.265	1.38	.171	.553	3.78	1.657	2.334	.022	.556		
		G2	.482	2.78	1.582			.519	3.64	1.588			.51	3.12	1.118			.503		
	FC	G1	.563	5.96	1.84	3.943	<.001	.551	6.22	2.902	1.522	.132	.557	5.82	1.976	3.095	.003	.557	G2	.507
		G2	.499	4.58	1.655			.517	5.46	2.012			.505	4.72	1.552			.507		
	FT	G1	.559	1201.28	429.081	2.304	.023	.562	1401.4	600.724	2.316	.023	.557	1167.04	391.332	2.068	.041	.559		
		G2	.512	993.44	472.063			.51	1158.18	436.286			.513	1006.96	382.775			.512		
Eco-Label	FFD	G1	.474	100.9	66.445	−3.86	<.001						.519	113.34	70.524	−3.284	<.001	.496	G1	.48
		G2	.549	153.84	70.639								.611	157.04	62.289			.58		
	RV	G1	.5	.58	.575	−1.633	.106						.464	.28	.497	−5.629	<.001	.482		
		G2	.564	.78	.648								.559	1	.756			.562		
	FC	G1	.486	1.42	.992	−2.268	.026						.475	1.06	.74	−5.854	<.001	.481	G2	.583
		G2	.542	1.86	.948								.612	2.1	1.015			.577		
	FT	G1	.462	163.92	110.665	−4.119	<.001						.461	143.9	94.325	−6.853	<.001	.461		
		G2	.584	261.56	125.918								.645	296.36	125.908			.615		
Brand	FFD	G1	.553	139.4	61.739	−.442	.66											.553	G1	.52
		G2	.565	144.92	63.266													.565		
	RV	G1	.505	1.06	.935	−.632	.529											.505		
		G2	.514	1.18	.962													.514		
	FC	G1	.509	2.22	1.234	−.914	.363											.509	G2	.537
		G2	.535	2.46	1.388													.535		
	FT	G1	.513	385.06	247.661	−.816	.417											.513		
		G2	.533	426.14	255.861													.533		

Table 2
DGRA and T-TEST comparison of AOIs among two groups in the study 2.

Product			Product D					Product E					Product F					Averages of Metrics	Averages of AOIs	
			DGRA	T-TEST				DGRA	T-TEST				DGRA	T-TEST				DGRA		
AOI	Metric	GRP	GRG	Mean	SD	T	p-v	GRG	Mean	SD	T	p-v	GRG	Mean	SD	T	p-v	GRG	GRP	GRG
Product Info	FFD	G1	.521	175.62	47.327	.627	.532	.498	171.00	48.201	−2.622	.010	.501	179.82	52.730	−.325	.746	.507	G1	.514
		G2	.512	169.62	48.422			.539	202.12	68.699			.518	183.86	70.240			.523		
	RV	G1	.516	3.50	1.216	.945	.347	.502	3.52	1.054	−3.263	.002	.551	3.88	1.380	−.861	.392	.523		
		G2	.518	3.26	1.322			.560	4.50	1.843			.551	4.12	1.409			.543		
	FC	G1	.505	5.54	1.809	.721	.473	.491	5.82	1.674	−2.967	.004	.538	6.32	1.889	−.271	.787	.512	G2	.531
		G2	.507	5.28	1.796			.551	7.10	2.550			.530	6.42	1.797			.529		
	FT	G1	.524	1390.34	498.887	.448	.655	.503	1414.76	455.832	−2.838	.006	.522	1553.10	553.479	−.459	.647	.516		
		G2	.496	1348.36	435.768			.566	1736.10	658.350			.530	1602.92	531.159			.531		
Price	FFD	G1	.533	155.62	43.203	1.935	.056	.547	133.48	38.445	1.557	.123	.525	147.82	54.067	2.086	.040	.535	G1	.529
		G2	.495	138.82	43.625			.505	122.56	31.325			.489	128.94	34.229			.496		
	RV	G1	.561	4.02	1.078	1.350	.180	.519	3.12	.773	1.124	.264	.529	3.5	1.035	1.659	.100	.536		
		G2	.542	3.72	1.144			.517	2.90	1.147			.513	3.14	1.246			.524		
	FC	G1	.532	5.80	1.471	1.829	.070	.513	5.10	1.313	1.279	.204	.514	5.36	1.481	1.047	.298	.520	G2	.507
		G2	.505	5.28	1.371			.503	4.74	1.496			.513	5.02	1.755			.507		
	FT	G1	.543	1238.68	351.816	2.767	.007	.515	998.46	300.165	1.104	.273	.517	1078.30	385.309	.984	.327	.525		
		G2	.498	1050.90	325.404			.504	924.80	364.154			.505	996.16	446.900			.502		
Eco-Label	FFD	G1						.494	99.94	69.953	−2.303	.012						.494	G1	.468
		G2						.532	130.12	60.780								.532		
	RV	G1						.462	.52	.677	−4.128	<.001						.462		
		G2						.576	1.20	.948								.576		
	FC	G1						.463	1.30	.995	−3.913	<.001						.463	G2	.579
		G2						.603	2.18	1.240								.603		
	FT	G1						.452	170.18	123.751	−3.911	<.001						.452		
		G2						.605	276.04	145.990								.605		
Brand	FFD	G1											.542	153.22	62.468	.996	.322	.542	G1	.544
		G2											.530	140.02	69.875			.530		
	RV	G1											.501	1.32	.891	−.483	.630	.501		
		G2											.533	1.42	1.162			.533		
	FC	G1											.587	2.72	1.386	.347	.730	.587	G2	.544
		G2											.575	2.62	1.497			.575		
	FT	G1											.547	501.30	289.727	.524	.602	.547		
		G2											.537	470.42	300.046			.537		

The DGRA method involves several key steps.

Step 1- Calculate Absolute Differences: The absolute difference $|\Delta(j)|$ between the sample and maximum value of all samples in each criterion is calculated for each sample by:

$$|\Delta_{0k}(j)| = |x_0(j) - x_k(j)| \tag{3}$$

In the above formula j r individual observations or participants in each group (1–50 per group). K also represents the products being compared, with values ranging from 1 to 6.

Step 2- Determine Grey Relational Variator: For each group, the average absolute difference $\Delta_{avg}(j)$ is calculated. Using this vector, the global grey relational variator ψ(j) is determined.

$$\Delta_{avg}(j) = \frac{1}{m}\sum_{k=1}^{m}|x_0(j) - x_k(j)| \tag{4}$$

$$\psi(j) = \frac{\Delta_{avg}(j)}{\max_k \max_j |x_0(j) - x_k(j)|} \tag{5}$$

Step 3- Estimate Distinguishing Coefficients (ξ(j)): Using ψ(j), the optimal value of multiplier h is estimated. The dynamic distinguishing coefficients ξ(j) are derived as $h \times \psi(j)$. For calculating the optimal value of the multiplier h is used. At first, a matrix $v(j)$ with h iterating from 1 to 2 in small steps, such that $v(j) = h \times \psi(j)$ is constructed. For each criterion j, the maximum value $maxv(j)$ from the column is chosen. It should not contain any value higher than 1. The value of h corresponding to the set $\{maxv(1), maxv(2), \ldots, maxv(n)\}$ is the approximately optimal h. N refers to the total observations or data points per criterion, corresponding to group size or product count.

Step 4- Calculate Grey Relational Coefficients γ(j): The grey relational coefficients γ(j) for each sample of groups are calculated using ξ(j) and |Δ(j)| using the following equation. m is the number of groups that now is 2 (Mahmoudi et al., 2020).

$$\gamma_{0k}(j) = \frac{\Delta_{\min} + \xi(j)_{0k}\,\Delta_{\max}}{|\Delta_{0k}(j)| + \xi(j)_{0k}\,\Delta_{\max}}, \xi(j) \in \left(0, 1\right], k = 1, 2, \cdots, m \tag{6}$$

Step 5- Determine Grey Relational Grades Γ: The weighted average of coefficients is computed to derive the grey relational grades Γ for each sample. Higher Γ indicates stronger relation or higher performance or higher attention. In this research, because the observations have the same weight, then the value of Γi (GRG) is the average of γ(j) (GRC).

$$\Gamma_{0k} = Average(\gamma_{0k}(j)) \tag{7}$$

Step 6- Calculate Grey Relational Standard Deviations σΓ (GRSD). N is the number of observations in this research equal to 50 for each group.

$$\sigma_{\Gamma_{0k}} = \sqrt{\frac{\sum_{j=1}^{n}(\Gamma_{0k} - \gamma_{0k}(j))^2}{n-1}} \tag{8}$$

Step 7- Calculation of Interval Grey Number Using GRG for each group of participants:

$$\text{GRG Interval} = \left[\Gamma_{0k} - \sigma_{\Gamma_{0k}}, \Gamma_{0k} + \sigma_{\Gamma_{0k}}\right] \tag{9}$$

Step 8 – Ranking and classification of variables based on GRG.

A chi-square analysis was performed to analyze the questionnaire data, carefully examining the necessary conditions. When these conditions were not met, Fisher's exact test served as an alternative approach.

4. Descriptive analysis

4.1. Part 1: eye-tracking data, DGRA and T-TEST results

Heatmap Analysis. To provide a qualitative perspective on participants' visual attention to the research variables, the heatmap analysis was employed before the quantitative analysis. Figs. 3 and 4 display heatmaps for the two participant groups, showing their visual attention toward products A, B, and C in Study 1 and products D, E, and F in Study 2. These heatmaps offer an effective means of showing the intensity of visual attention. Focusing on AOI helps to interpret fixation data more intuitively (Kulkarni and Yuan, 2015). Heatmaps use a gradient of colors to display fixation intensity, with red indicating the highest attention, yellow showing moderate focus, and green representing the least fixation. Areas that appear without color indicate no visual engagement. This visual representation is useful for identifying attention patterns and highlighting key areas of interest (He et al., 2024).

In Fig. 3 (Study 1), the first section represents the control group (G1), demonstrating high visual attention to the price factor across all three products. Significant attention is directed toward the product images, particularly for Products A and B. In contrast, the second section illustrates the visual attention patterns of Group G2, participants exposed to an environmental educational video message. This group shows an increased focus on the eco-friendly information associated with Products A and C, along with greater attention to brand information for Product A. However, there is a noticeable decrease in attention to the price factor among participants in Group G2.

Overall, participants exposed to an environmental educational video message exhibited a shift in attention from the price factor to eco-friendly labels and information related to Products A and C, compared to the control group. This shift was especially significant for Product A, a well-known brand, highlighting the positive influence of a familiar eco-friendly brand in enhancing attention to eco-friendly information. These findings suggest that exposure to an environmental educational video message positively impacts participants' perception and attention to eco-friendly information on the product display page.

In Fig. 4 (Study 2), the first section, which represents Group G1, shows high visual attention to price across all three products. This is especially noticeable for Product D, which is neither branded nor eco-friendly and has the lowest price among the options. Additionally, this group shows significant attention to brand cues for Product F, a familiar brand.

In the second section, representing Group G2, there is a noticeable reduction in attention to price, accompanied by an increased focus on product information. Specifically, this group pays heightened attention to the eco-friendly features associated with Product E. Conversely, there is a decline in visual attention to brand cues for Product F, which, although familiar, is not an eco-friendly brand.

Overall, participants exposed to an environmental educational video message showed reduced attention to the price factor and increased attention to product information compared to the control group. A significant increase in attention to eco-friendly cues was observed for the eco-friendly product. The findings align with those of Study 1, with an important distinction: when environmental awareness was heightened, visual attention to brand cues decreased for a familiar but non-eco-friendly product.

4.2. DGRA and T-TEST

AOI Product Info Study 1. For FFD metric, Group G2 exhibited significantly longer FFDs than Group G1 for Product A ($GRG = .609$, $Mean = 271.56$, $T = -2.281$, $p = 0.025$) and Product C ($GRG = .605$, $Mean = 248.34$, $T = -2.277$, $p = 0.007$), indicating greater initial



Fig. 3. Heatmaps of products A, B, and C (Groups: G1 and G2) Study 1.



Fig. 4. Heatmaps of products D, E, and F (Groups: G1 and G2) Study 2.

attention to the Product Infos. Conversely, for Product B, Group G1 had a significantly longer FFD than Group G2 (*GRG*:.557, *Mean* = 222.96, $T = 2.186$, $p = 0.031$). Regarding the RV metric, Group G2 showed a higher mean RV for Products A (*GRG* = .519, *Mean* = 3.36, $T = -1.465$, $p = 0.146$) and C (*GRG* = .525, *Mean* = 3.04, $T = -1.696$, $p = 0.093$), while Group G1 showed a higher mean RV for Product B (*GRG* = .518, *Mean* = 2.62, $T = 1.533$, $p = 0.129$). For the FC metric, Group G2 demonstrated significantly higher counts for Product A (*GRG* = .527, *Mean* = 7.34, $T = -2.764$, $p = 0.007$) and Product C (*GRG* = .534, *Mean* = 6.3, $T = -2.802$, $p = 0.006$), indicating more frequent engagement with the eco-friendly Product Infos, while that showed higher in G1 for Product B (*GRG* = .522, *Mean* = 5.1, $T = 1.405$, $p = 0.163$). In terms of the FT metric, Group G2 showed a significantly longer FT for Products A (*GRG* = .539, *Mean* = 2709.16, $T = -1.981$, $p = 0.051$) and C (*GRG* = .552, *Mean* = 2477.36, $T = -2.163$, $p = 0.033$), reflecting more sustained attention, whereas Group G1 exhibited a longer FT for Product B (*GRG* = .536, *Mean* = 2154.9, $T = 1.988$, $p = 0.050$).

AOI Price Study 1. For the FFD metric, Group G1 exhibited longer

FFDs than Group G2 across all products. When comparing products, Product B (*GRG* = .569, *Mean* = 170.76, $T = 2.614$, $p = 0.010$) showed greater initial attention to price information within this group compared to Product A (*GRG* = .560, *Mean* = 149.16, $T = 1.305$, $p = 0.195$) and C (*GRG* = .531, *Mean* = 147.08, $T = 1.056$, $p = 0.294$). For the RV, similar to the FFD metric, Group G1 exhibited significantly higher RVs than Group G2 across all products. When comparing the products, Product B (*GRG* = .550, *Mean* = 4.18, $T = 1.380$, $p = 0.171$) showed more frequent returns to price information compared to Product A (*GRG* = .563, *Mean* = 4, $T = 3.943$, $p < 0.001$) and Product C (*GRG* = .553, *Mean* = 3.78, $T = 2.334$, $p = 0.022$). For the FC metric, Group G1 exhibited significantly higher fixation counts across all products than Group G2. Among the products, Product B (*GRG* = .551, *Mean* = 6.22, $T = 1.522$, $p = 0.132$) reflected more frequent fixations on price-related areas compared to Product A (*GRG* = .563, *Mean* = 5.96, $T = 3.943$, $p < 0.001$) and Product C (*GRG* = .557, *Mean* = 5.82, $T = 3.095$, $p = 0.003$). Finally, for the FT metric, Group G1 consistently demonstrated longer fixation times on price information across all three products. Significant differences were observed for Product B (*GRG* = .562, *Mean* = 1401.4, $T = 2.316$, $p = 0.023$), which had higher mean values compared to Product A (*GRG* = .559, *Mean* = 1201.28, $T = 2.304$, $p = 0.023$), and Product C (*GRG* = .557, *Mean* = 1167.04, $T = 2.068$, $p = 0.041$).

AOI Eco-Label Study 1. For the FFD metric, Group G2 exhibited significantly longer durations compared to Group G1 for both Product A (*GRG* = .549, *Mean* = 153.84, $T = -3.860$, $p < 0.001$) and Product C (*GRG* = .611, *Mean* = 157.04, $T = -3.284$, $p < 0.001$), suggesting greater initial attention to eco-label information among Group G2. Regarding the RV metric, same as FFD, Group G2 displayed a greater mean than Group G1 for both Product A (*GRG* = .564, *Mean* = .780, $T = -1.633$, $p = 0.106$) and Product C (*GRG* = .559, *Mean* = 1, $T = -5.629$, $p < 0.001$). For the FC metric, Group G2 had significantly higher fixation counts than Group G1 for both Product A (*GRG* = .542, *Mean* = 1.86, $T = -2.268$, $p = 0.026$) and Product C (*GRG* = .612, *Mean* = 2.1, $T = -5.854$, $p < 0.001$), reflecting more frequent fixations on eco-labels in the video-exposed group. In terms of the FT metric, Group G2 consistently demonstrated significantly longer fixation times compared to Group G1 for both Product A (*GRG* = .584, *Mean* = 261.56, $T = -4.119$, $p < 0.001$) and Product C (*GRG* = .645, *Mean* = 296.36, $T = -6.853$, $p < 0.001$), highlighting more sustained attention to eco-label information in this group.

AOI Brand Study 1. For the FFD metric, Group G2 showed a slightly higher mean than Group G1, suggesting a minor increase in initial attention to brand-related information (*GRG* = .565, *Mean* = 144.92, $T = -.442$, $p = 0.660$). Similarly, for the RV metric, the mean was slightly higher in Group G2 compared to Group G1, indicating a modest tendency for more revisits to brand-related areas (*GRG* = .514, *Mean* = 1.18, $T = -.632$, $p = 0.529$). Regarding the FC metric, Group G2 had a slightly higher mean than Group G1, reflecting a potential increase in fixation frequency (*GRG* = .535, *Mean* = 2.46, $T = -.914$, $p = 0.363$). Finally, for the FT metric, Group G2 exhibited a longer mean fixation time than Group G1, indicating more sustained attention to brand-related information (*GRG* = .533, *Mean* = 426.14, $T = -.816$, $p = 0.417$). Overall, Group G2 demonstrated a higher level of visual attention to the Product Info, Eco-Label, and Brand (*GRG* = .532; .583; .537), while Group G1 showed a higher level of attention to the Price factor (*GRG* = .556).

AOI Product Info Study 2. For the FFD metric, Group G1 had a higher mean than Group G2 for Product D (*GRG* = .521, *Mean* = 175.62, $T = .627$, $p = 0.532$). In contrast, Group G2 exhibited a greater mean for Product E (*GRG* = .539, *Mean* = 202.12, $T = -2.622$, $p = 0.010$) and Product F (*GRG* = .518, *Mean* = 183.86, $T = -.325$, $p = 0.746$). Among all products, Product E in Group G2 recorded the highest mean for this metric. For the RV metric, differences between groups for Product D were minimal. Group G2 had higher means for Product E (*GRG* = .560, *Mean* = 4.5, $T = -3.263$, $p = 0.002$) and Product F (*GRG* = .551, *Mean* = 4.12, $T = -.861$, $p = 0.392$). Notably, Group G2 had significantly more revisits for Product E, which recorded the highest frequency of returns to product descriptions among all products. Product D and Product F demonstrated minimal differences in FC metric values across groups. In contrast, Group G2 demonstrated significantly higher means for Product E (*GRG* = .551, *Mean* = 7.10, $T = -2.967$, $p = 0.004$), with the most frequent fixations observed for this product across all groups. Regarding the FT metric, Group G1 had a greater mean than Group G2 for Product D (*GRG* = .524, *Mean* = 1390.34, $T = .448$, $p = 0.655$). However, Group G2 exhibited significantly longer means for Product E (*GRG* = .566, *Mean* = 1736.10, $T = -2.838$, $p = 0.006$) and Product F (*GRG* = .530, *Mean* = 1602.92, $T = -.459$, $p = 0.647$). Overall, Product E in Group G2 recorded the highest mean for this metric across all products.

AOI Price Study 2. For the FFD metric, Group G1 exhibited longer FFDs than Group G2 across all products. When comparing products, Product D (*GRG* = .533, *Mean* = 155.62, $T = 1.935$, $p = 0.056$) showed greater initial attention to price information within this group compared to Product F (*GRG* = .525, *Mean* = 147.82, $T = 2.086$, $p = 0.040$) and E (*GRG* = .547, *Mean* = 133.48, $T = 1.557$, $p = 0.123$). For the RV, similar to the FFD metric, Group G1 exhibited significantly higher RVs than Group G2 across all products. When comparing the products, Product D (*GRG* = .561, *Mean* = 4.02, $T = 1.350$, $p = 0.180$) and Product F (*GRG* = .529, *Mean* = 3.5, $T = 1.659$, $p = 0.100$) showed more frequent returns to price information compared to Product E (*GRG* = .519, *Mean* = 3.12, $T = 1.124$, $p = 0.264$). For the FC metric, Group G1 exhibited significantly higher fixation counts across all products than Group G2. Among the products, Product D (*GRG* = .532, *Mean* = 5.80, $T = 1.829$, $p = 0.070$) reflected more frequent fixations on price-related areas compared to Product F (*GRG* = .514, *Mean* = 5.36, $T = 1.047$, $p = 0.298$) and Product E (*GRG* = .513, *Mean* = 5.10, $T = 1.279$, $p = 0.204$). Finally, for the FT metric, Group G1 consistently demonstrated longer fixation times on price information across all three products. Significant differences were observed for Product D (*GRG* = .543, *Mean* = 1238.68, $T = 2.767$, $p = 0.007$), which had higher mean values compared to Product F (*GRG* = .517, *Mean* = 1078.30, $T = .984$, $p = 0.327$) and Product E (*GRG* = .515, *Mean* = 998.46, $T = 1.104$, $p = 0.273$).

AOI Eco-Label Study 2. For the FFD metric, Group G2 demonstrated a significantly longer duration (*GRG* = .532, *Mean* = 130.12, $T = -2.303$, $p = 0.012$), suggesting greater initial attention to the eco-label among Group G2. Regarding the RV metric, Group G2 exhibited a significantly higher mean (*GRG* = .576, *Mean* = 1.2, $T = -4.128$, $p < 0.001$), indicating a stronger tendency to revisit the eco-label than Group G1. For the FC metric, Group G2 also showed significantly more fixations (*GRG* = .603, *Mean* = 2.18, $T = -3.913$, $p < 0.001$), reflecting more frequent engagement with the eco-label. Regarding the FT metric, Group G2 demonstrated significantly longer fixation times on the eco-label (*GRG* = .605, *Mean* = 276.04, $T = -3.911$, $p < 0.001$), highlighting more sustained attention.

AOI Brand Study 2. Group G1 had a slightly higher mean than Group G2 for the FFD metric (*GRG* = .542, *Mean* = 153.22, $T = .996$, $p = 0.322$). Regarding the RV metric, Group G2 showed a slightly higher mean (*GRG* = .533, *Mean* = 1.42, $T = -.483$, $p = 0.630$). The FC metric's mean was slightly higher in Group G1 compared to Group G2 (*GRG* = .587, *Mean* = 2.82, $T = .347$, $p = 0.730$). Finally, for the FT metric, Group G1 exhibited a slightly longer mean fixation time than Group G2 (*GRG* = .547, *Mean* = 501.30, $T = .524$, $p = 0.602$). Overall, Group G2 demonstrated a higher visual attention to the Product Info and Eco-Label (*GRG* = .531; .579), while Group G1 showed higher attention to the Price factor (*GRG* = .529). GRG values for the Brand factor were closely aligned between groups. However, Group G1 tended to exhibit slightly greater visual attention to this factor.

4.3. Part 2: questionnaire data

Following the recommendations of McShane et al. (2024), statistical results are interpreted by considering both p-values and test statistics (F) rather than relying solely on significance thresholds (McShane et al.,

2024). In this study, 'F' represents test statistics value derived from the Fisher-Freeman-Halton Exact Test and Pearson Chi-Square Test, providing additional insight into the association strength between categorical variables.

Chi-square analysis and Fisher-Freeman-Halton Exact Test were employed to examine demographic characteristics and responses across intervention groups. Results indicate a high similarity between G1 and G2 across key demographic factors. Age distribution was relatively close between groups ($F = .918$, $p = 0.851$), with participants distributed within the age range of 18–29 years. An equal distribution of men and women is observed in both groups ($F = .040$, $p = 1$). Most participants were single across both groups ($F = .543$, $p = 0.715$). Participants were similarly distributed across Bachelor's, Master's, and Doctorate levels ($F = .456$, $p = 0.855$).

In Study 1, a significant difference in product choices was observed ($F = 7.785$, $p = 0.019$). Participants in Group G1 showed a stronger preference for Products A and B, whereas those in Group G2 displayed a greater inclination toward Products A and C. Although the p-value did not meet widely used thresholds for inferential claims, the magnitude of the test statistic ($F = 5.98$, $p = 0.189$) suggests a potentially meaningful trend toward greater price sensitivity in Group G1. Importance ratings for product explanations were similar across groups ($F = 4.688$, $p = 0.184$), and brand influence was moderately to highly rated by both groups, with differences observed in mean scores ($F = 3.025$, $p = 0.563$).

In Study 2, product choices differed between groups ($F = 10.832$, $p = 0.004$). Group G1 demonstrated a stronger preference for Products D and F, whereas Group G2 exhibited a noticeably greater preference for Product E. Consistent with Study 1, Group G1 exhibited greater price sensitivity ($F = 5.893$, $p = 0.180$), with the test statistic indicating a meaningful directional trend. Participants from both groups rated product explanation importance similarly ($F = 3.169$, $p = 0.536$). Brand influence ratings differed between groups ($F = 6.081$, $p = 0.193$), with Group G1 assigning relatively greater importance to this factor.

G2 demonstrated a strong emotional reaction to the video message, with 62 % reporting being "extremely affected" and 36 % "very affected". Group G2 demonstrated a significantly higher understanding of the products' eco-friendly information (96 %) than Group G1 (74 %; $F = 9.490$, $p = 0.004$). Significant differences were observed in pre-existing environmental awareness ($F = 16.916$, $p < 0.001$), with Group G2 demonstrating higher levels of complete awareness than Group G1. Trust in eco-friendly information was slightly higher in Group G2 ($F = 7.874$, $p = 0.072$), suggesting a possible trend, although the p-value falls outside commonly used decision thresholds. In terms of willingness to purchase eco-friendly products, Group G2 displayed a greater inclination toward green purchases ($F = 7.441$, $p = 0.111$). Similarly, willingness to pay more for eco-friendly products was higher in Group G2 ($F = 7.971$, $p = 0.085$), reinforcing the broader behavioral shift observed in response to the intervention. The results are presented in Table 3.

Table 3
Demographic and questionnaire results.

Variables	Categories	G1		G2		F	p-v
		n	%	n	%		
Age group	15–17	0	0	0	0	.918	.851
	18–20	14	28	12	24		
	21–23	18	36	19	38		
	24–26	10	20	8	16		
	27–29	8	16	11	22		
Gender	Male	25	50	24	48	.040	1
	Female	25	50	26	52		
Marital status	Single	47	94	45	90	.543	.715
	Married	3	6	5	10		
Education level	Bachelor's	25	50	27	54	.456	.855
	Master's	18	36	15	30		
	Doctorate	7	14	8	16		
Product Choice study 1	Product A	21	42	22	44	7.785	.019
	Product B	19	38	8	16		
	Product C	10	20	20	40		
Price study 1	Not at all important	1	2	2	4	5.98	.189
	Slightly important	5	10	11	22		
	Moderately important	15	30	19	38		
	Very important	20	40	14	28		
	Extremely important	9	18	4	8		
Explanation study 1	Not at all important	0	0	0	0	4.688	.184
	Slightly important	5	10	2	4		
	Moderately important	17	34	10	20		
	Very important	27	54	36	72		
	Extremely important	1	2	2	4		
Brand study 1	Not at all influential	1	2	1	2	3.025	.563
	Slightly influential	2	4	6	12		
	Moderately influential	14	28	16	32		
	Very influential	23	46	20	40		
	Extremely influential	10	20	7	14		
Product Choice study 2	Product D	20	40	10	20	10.832	.004
	Product E	9	18	24	48		
	Product F	21	42	16	32		
Price study 2	Not at all important	0	0	2	4	5.893	.180
	Slightly important	4	8	9	18		
	Moderately important	18	36	21	42		
	Very important	22	44	14	28		
	Extremely important	6	12	4	8		
Explanation study 2	Not at all important	3	6	1	2	3.169	.536
	Slightly important	3	6	4	8		
	Moderately important	18	36	12	24		
	Very important	20	40	25	50		
	Extremely important	6	12	8	16		
Brand study 2	Not at all influential	2	4	4	8	6.081	.193
	Slightly influential	5	10	7	14		

(continued on next page)

5. Hypothesis testing

This section presents the results of all hypothesis tests, highlighting how empirical evidence supports or contradicts each hypothesis while emphasizing statistical evidence and the implications of the findings.

H1. Exposure to environmental educational messages right before the product selection phase triggers emotional arousal, which enhances visual attention to eco-friendly information and eco-labels.

This hypothesis was supported and showed strong statistical evidence. The results from the eye-tracking device revealed a significant increase in visual attention to eco-friendly product attributes, including eco-friendly information and eco-labels, in Group G2, which was exposed to an environmental educational video message. The DGRA and T-test analyses from Study 1 and Study 2 indicated that Group G2 exhibited higher mean values across all eye-tracking metrics related to

Table 3 (*continued*)

Variables	Categories	G1		G2		F	p-v
		n	%	n	%		
	Moderately influential	8	16	16	32		
	Very influential	20	40	13	26		
	Extremely influential	15	30	10	20		
The effect of the video on emotions	Not at all affected			0	0		
	Slightly affected			0	0		
	Moderately affected			1	2		
	Very affected			18	36		
	Extremely affected			31	62		
Eco-friendly info	No	13	26	2	4	9.490	.004
	Yes	37	74	48	96		
Pre-existing environmental awareness	No awareness at all	2	4	0	0	16.916	<.001
	Somewhat no awareness	11	22	2	4		
	Neutral	17	34	12	24		
	Somewhat aware	18	36	24	48		
	Completely aware	2	4	12	24		
Trust to Eco-friendly info	Do not trust at all	1	2	0	0	7.874	.072
	Somewhat do not trust	13	26	6	12		
	Neutral	14	28	11	22		
	Somewhat trust	20	40	25	50		
	Completely trust	2	4	8	16		
Eco-friendly purchasing	Not at all important	3	6	2	4	7.441	.111
	Slightly important	13	26	6	12		
	Moderately important	19	38	14	28		
	Very important	13	26	24	48		
	Extremely important	2	4	4	8		
Eco-friendly price	Not at all likely	4	8	0	0	7.971	.085
	Slightly likely	9	18	7	14		
	Neutral	17	34	11	22		
	Very likely	7	14	11	22		
	Extremely likely	13	26	21	42		

eco-friendliness compared to Group G1. This finding was further supported by the heat maps (Figs. 3 and 4), which showed increased focus on areas related to the eco-friendly aspects of the products. Additionally, the results of the questionnaire highlighted a notable emotional impact of the video on the participants.

These findings align with the ELM, particularly the peripheral route of persuasion. As per ELM, exposure to emotional cues (in this case, environmental video messages) engages peripheral processing by evoking emotional responses that drive attention to product attributes, such as eco-friendly information and eco-labels (Luo et al., 2024; Mosler and Martens, 2008; S. Zhang et al., 2024). Furthermore, the results align with the Priming Effect, where the timing of environmental stimuli right before product selection activates subconscious attitudes and enhances visual attention to eco-friendly information, confirming the theory's suggestion that exposure to stimuli impacts subsequent attention and decision-making (Lindblom, 2023; Tate et al., 2014; Zsidó, 2024).

H2. Increased visual attention to eco-friendly information significantly enhances consumers' preference for sustainable products over non-eco-friendly products, regardless of brand influence.

This hypothesis was supported and showed strong statistical evidence. Participants in Group G2 displayed increased visual attention to eco-friendly factors, as indicated by the eye-tracking data analysis from the DGRA and the T-test results. Additionally, the questionnaire findings revealed that these participants had significantly higher levels of understanding and trust regarding the products' eco-friendly information. Furthermore, the final product selection in this group showed a significant shift toward eco-friendly products in both studies. In Study 1, Group G2 exhibited a stronger preference for Products A and C, both of which were eco-friendly. In Study 2, this group showed a greater inclination toward Product E, the only eco-friendly option among the three products.

These findings confirm the relevance of ELM's central route processing, where increased attention to eco-friendly information leads to greater cognitive engagement with the product's environmental benefits, thereby shaping attitudes and influencing purchase intentions (Luo et al., 2024; Petty et al., 1997). In alignment with the Priming Effect, exposure to the video message primed participants' environmental awareness, guiding them to focus on green product features, which then increased their preference for eco-friendly options (Lindblom, 2023; Zsidó, 2024). This supports the theory's claim that priming can influence both visual attention and decision-making.

H3. Exposure to environmental educational messages right before the product selection reduces price sensitivity by moderating visual attention.

This hypothesis was supported and showed strong statistical evidence. In both studies (Study 1 and Study 2), the heat maps (Figs. 3 and 4) and the results from the DGRA and T-test analyses of eye-tracking data revealed a significant decrease in visual attention to the price factor for all products among participants in Group G2. Furthermore, the questionnaire results indicated that this group exhibited a greater willingness to pay more for eco-friendly products.

These findings are grounded in ELM's central route processing and the Priming Effect, where increased environmental awareness reduces the cognitive weight of price and shifts attention toward environmentally relevant product attributes. This also aligns with previous research suggesting that priming environmental concerns leads to lower price sensitivity and a greater willingness to pay for sustainable products (Bimonte et al., 2020; Luo et al., 2024; Marroquín-Ciendúa et al., 2020; Tate et al., 2014).

H4a. When the product is eco-friendly, environmental awareness strengthens the positive impact of a well-known brand on consumer preference.

This hypothesis was supported and showed strong statistical evidence. Study 1 revealed that in Group G2, who were exposed to increased environmental awareness through an environmental educational video message, Product A, as a familiar and eco-friendly brand, exhibited higher mean values across all eye-tracking metrics (except for mean price sensitivity, which negatively influences product choice) compared to both Product B and even Product C, an eco-friendly product without a brand. These findings, based on the eye-tracking metrics (average) analyses presented in Table 1, were further supported by the questionnaire results, which indicated a stronger preference for Product A over the other two products.

This finding is consistent with TPB, which posits that high environmental awareness enhances positive attitudes toward green products, especially when the product is from a well-known brand. The increased environmental awareness likely enhances subjective norms, leading participants to place greater value on eco-friendly brands (Eid et al., 2021; Ewe and Tjiptono, 2023; García-Salirrosas et al., 2024). This suggests that brand familiarity can have a stronger influence on

consumer preferences when it is aligned with high environmental awareness.

H4b. When the product is non-eco-friendly, environmental awareness weakens the impact of a well-known brand on consumer preference.

This hypothesis was supported and showed strong statistical evidence. In Study 2, it was found that Group G2 exhibited less visual attention to Product F, a familiar but non-eco-friendly brand, across all eye-tracking metrics compared to Product E, an eco-friendly but unbranded product. Based on the DGRA and T-test findings presented in Table 2, the results indicated that Product E captured greater attention in all metrics. Moreover, a comparison between Group G2 and Group G1 revealed a decrease in visual attention to brand-related factors in Group G2. Questionnaire results also supported these findings, indicating that Group G2 showed a significantly stronger preference for Product E and reported that the brand had a reduced influence on their final choice compared to Group G1.

This finding aligns with the TPB. In this case, the high environmental awareness among Group G2 likely diminished the influence of brand familiarity when faced with a non-eco-friendly product, as their attitudes toward environmental sustainability prioritized eco-friendly attributes over brand familiarity. This is consistent with this theory's based claim that heightened environmental awareness can diminish the influence of brand familiarity when consumers are presented with a non-eco-friendly brand (Ewe and Tjiptono, 2023).

H5. Under conditions of heightened environmental awareness, consumers show stronger engagement with eco-friendly products from familiar brands compared to non-branded alternatives.

This hypothesis was supported and showed strong statistical evidence. In Study 1, as shown in Hypothesis H4a within Group G2, Product A had the highest average across all eye-tracking metrics, except for mean price sensitivity. These findings persisted even when compared to Product C, an eco-friendly but non-branded product. Additionally, the questionnaire findings indicated that while this group exhibited an increased tendency to choose both eco-friendly products, their preference for Product A, as a familiar brand, was notably stronger. This supports previous research suggesting that familiar brands have significant communication advantages and attract more consumer attention than unfamiliar brands (Belloque et al., 2023; Delgado-Ballester et al., 2012; Oanh, 2024).

H6. Brand influence reduces consumers' price sensitivity, as evidenced by decreased visual attention to the price factor when encountering a familiar brand.

This hypothesis lacked support and was deemed insignificant. Although increasing environmental awareness, as demonstrated in the previous hypotheses, resulted in a decrease in attention to the price factor across all products, including branded products, in both studies, a different perspective emerges when considering the brand factor and comparing the mean visual attention to the price factor between products within each group and across both studies. Based on the DGRA and T-test results, the following findings emerged: In Study 1, the eye-tracking data showed that visual attention to Products A and C was relatively similar and generally closely aligned. In Study 2, within Group G1, the lowest visual attention to price was noted for Product E, a non-branded item. A similar trend was observed in Group G2; however, the visual attention for Product E was closer to that of Product F, which is a familiar brand product. Overall, across both studies, the product with the lowest price consistently attracted the highest visual attention to the price factor.

6. General discussion and conclusion

This study examined the role of environmental educational video messages in enhancing visual attention to eco-friendly information and influencing green product preferences among Generation Z consumers on online shopping platforms. By combining eye-tracking technology with a questionnaire, the research explored how environmental awareness and brand familiarity interact to shape sustainable consumption behaviors. A key contribution of this research is the analysis of the timing of environmental educational video messages, particularly their impact when presented right before the product selection phase. This design highlights situations where participants do not have enough time for deep, long-term cognitive processing, thereby emphasizing the immediate effects of environmental video messages on consumer attention and behavior.

6.1. Impact of environmental educational video message timing on green preferences

Consistent with the ELM and the Priming Effect, exposure to environmental educational video messages at a critical moment before decision-making can trigger emotional arousal. This, in turn, increases visual attention to eco-friendly information and eco-labels. In such cases based on the priming effect, exposure to an initial cue subconsciously influences consumer behavior in subsequent stages. This process is often driven by emotional arousal, which enhances visual attention, especially when the presented stimuli align with the initial cue (Lindblom, 2023; Zsidó, 2024). This momentary exposure significantly increased fixation durations, fixation counts, and revisits for eco-friendly cues, as observed in the eye-tracking data. Overall, a consistent pattern of preference for eco-friendly products emerged among participants exposed to environmental educational messages. In both studies, Group G2, which was exposed to an environmental educational video message, showed notably greater visual attention to eco-friendly products. This heightened attention led to a stronger inclination to choose products with eco-friendly attributes over non-eco-friendly options. These findings suggest that momentary exposure to these messages effectively highlights sustainability features during decision-making, influencing consumer choices even in fast-paced, time-constrained situations. These results highlight the immediate cognitive and emotional priming effect of environmental messages, which affect consumer attention and decision-making, even within limited timeframes. These findings align with previous research, demonstrating that decision-making often occurs rapidly and unconsciously when there is insufficient time for deep analysis (Kahneman and Egan, 2011).

6.2. Dual role of brand familiarity and environmental awareness

Based on TPB, brand familiarity exhibited a dual role under heightened environmental awareness:

Branded Eco-Friendly Product. When the branded product was eco-friendly (Study 1, Product A), environmental awareness enhanced the positive effect of brand familiarity on consumer preference. This was evidenced by the highest levels of visual attention recorded across all eye-tracking metrics. This finding supports previous research, which suggests that brand familiarity increases preference for sustainable products and positively influences consumer decisions when paired with eco-friendly attributes (Belloque et al., 2023; Ewe and Tjiptono, 2023; Yang, 2017). Additionally, when comparing Product A with Product C (eco-friendly but non-branded), participants in Group G2 displayed significantly greater visual attention toward Product A. This finding highlights the synergistic effect of eco-friendliness and brand familiarity, indicating that the presence of a well-known brand enhances consumer focus on sustainability attributes. This finding is consistent with prior research, showing that familiar brands among eco-friendly products more effectively capture consumer attention and enhance stronger engagement with sustainability attributes (Belloque et al., 2023; Delgado-Ballester et al., 2012; Oanh, 2024). The environmental educational video message effectively increased participants' attention to eco-friendly cues, and coupled with brand familiarity, Product A

emerged as the most visually engaging option. This greater tendency was also shown in the questionnaire results on product choice, which indicated a higher preference for Product A among participants in Group G2.

Branded Non-Eco-Friendly Product. In contrast, when the product lacked eco-friendly attributes (Study 2, Product F), brand familiarity lost its impact among participants with heightened environmental awareness. Participants in Group G2 showed significantly less visual attention to Product F (a familiar brand but non-eco-friendly product) and instead focused more on Product E (an eco-friendly but non-branded product). This indicates that increased environmental awareness, triggered by the environmental educational video message, can diminish the effect of brand familiarity when it conflicts with sustainability values. These findings align with prior research, which suggests that individuals with heightened environmental awareness not only perceive greater control over their choices to exhibit green behavior but also become less influenced by brand familiarity when faced with non-eco-friendly products (Choi and Johnson, 2019; Ewe and Tjiptono, 2023).

6.3. Reduction of price sensitivity

The results indicated a significant reduction in price sensitivity after participants were exposed to the environmental educational video message. Participants in Group G2 showed decreased visual attention to price cues in both studies, especially as demonstrated by shorter fixation durations and fewer revisits to price information. This change reflects a reorientation of consumer priorities, where environmental awareness influences the emphasis on price considerations. Notably, this reduction occurred even in situations that allowed limited time for deep cognitive processing, highlighting the immediate and short-term impact of environmental messages. These findings are consistent with previous research, which emphasizes that environmental awareness encourages consumers to prioritize sustainable attributes over pricing and enhances their willingness to pay for eco-friendly products (Alhamad et al., 2023; Ansu-Mensah, 2021; Minbashrazgah et al., 2017).

6.4. Theoretical contributions

This study explores the timing of environmental educational video messages and their immediate impact on consumer decision-making regarding green products. It demonstrates that brief, targeted exposure to environmental cues right before decision-making can significantly influence consumer attention and choices (Zsidó, 2024). This finding contributes to the literature on the ELM and the Priming Effect by highlighting the effectiveness of momentary interventions in time-constrained purchasing scenarios, where cognitive and affective processes interact to shape decisions. This extends previous studies on digital consumer behavior, reinforcing the role of priming in sustainability marketing and consumer psychology.

Additionally, based on the TPB, the study underscores the dual role of brand familiarity in situations with high environmental awareness. When an eco-friendly product is associated with a well-known brand, brand familiarity strengthens consumer engagement and preference. Conversely, when a non-eco-friendly product carries a familiar brand name, environmental awareness reduces brand influence, prompting consumers to shift their preference toward eco-friendly but non-branded alternatives. This nuanced interaction between brand familiarity and environmental awareness provides a deeper understanding of consumer decision-making in sustainability contexts (Belloque et al., 2023; Ewe and Tjiptono, 2023; Yang, 2017).

Furthermore, the findings demonstrate that environmental awareness, triggered through environmental educational video messages, reduces visual attention to price information, leading to a reorientation of consumer priorities toward sustainability attributes. This insight contributes to the behavioral pricing literature by showing how environmental cues can mitigate price sensitivity, encouraging greater willingness to pay for green products. These insights align with previous research suggesting that environmental awareness influences consumer valuation of sustainability, even in the presence of competing economic factors (Alhamad et al., 2023).

Finally, by integrating eye-tracking technology with traditional questionnaires, this study offers a strong methodological framework for understanding the subconscious and explicit factors that drive green consumer behavior. By combining eye-tracking metrics with self-reported preferences, the research enhances the reliability of its findings. This study employed DGRA and T-test methods to analyze eye-tracking data, leveraging their combined strengths to address limitations of traditional approaches, such as challenges with statistical significance. Additionally, the application of DGRA presents a novel approach for analyzing eye-tracking data, particularly in cases where traditional significance testing using p-values may obscure meaningful insights. Recent discussions emphasize that p-values alone are insufficient for determining statistical significance; instead, best practices recommend incorporating complementary methods for a more comprehensive interpretation of results (Imbens, 2021; McShane et al., 2024; Rafi and Greenland, 2020). While some T-test results did not fall below the conventional p-value threshold for statistical significance in this study, the DGRA analysis effectively addressed these gaps. By incorporating DGRA alongside T-tests, this study ensures a multi-faceted interpretation of data, mitigating the risks of misinterpretation associated with sole reliance on p-values and reinforcing the robustness of the findings.

Overall, this study expands theoretical frameworks in consumer decision-making by demonstrating the priming effects of environmental messages, the complex interaction between brand familiarity and sustainability perceptions, and the behavioral impact of environmental cues on price sensitivity. It also presents a methodological advancement by integrating neuromarketing techniques and analytical methods to better capture implicit cognitive and emotional responses in green purchasing contexts.

6.5. Managerial implications

This study offers several practical recommendations for marketers, policymakers, and organizations aiming to promote sustainable consumption. The findings underscore the effectiveness of momentary exposure to an environmental educational video message in influencing consumer attention and choices. Policymakers can strategically place short, impactful educational video messages right before the product search results phase on online shopping platforms to enhance consumer focus on eco-friendly attributes and drive engagement with sustainable products.

For brands with an established market presence, aligning product attributes with sustainability can significantly boost consumer engagement. Companies with strong brand recognition should prioritize highlighting their eco-friendly features in marketing campaigns. This is especially effective when combining brand familiarity with environmental consciousness, which has been shown to enhance consumer attention and preferences. Given current policies aimed at increasing environmental awareness, brands must not only adopt environmental protection strategies but also enhance the sustainability of their products to maintain competitiveness in the market.

The decrease in price sensitivity among individuals exposed to the environmental educational video message indicates that awareness campaigns can help reduce the higher price barriers to adopting green products. Policymakers and businesses should prioritize more educational initiatives that highlight the environmental and social benefits of green products to encourage consumers to look beyond price considerations.

Based on these findings, online green retailers should improve the presentation of eco-friendly products by prominently displaying eco-labels and sustainability information on product pages. Optimizing the

placement of these attributes can attract consumers' visual attention, encouraging them to engage with the products more actively. Generation Z, known for its digital savviness and heightened environmental awareness, is a crucial target audience for sustainable marketing strategies. Therefore, businesses should tailor their messaging and platform design to meet this group's preferences for authenticity, social responsibility, and easy access to information. For example, platforms should integrate clear, concise environmental messaging directly on the homepage or in prominent product placement areas, ensuring that information is easily accessible.

Finally, the increasing use of neuromarketing techniques, including eye-tracking technology, highlights their potential to reveal subconscious consumer behaviors. These insights can assist organizations in refining their marketing strategies by ensuring that important sustainability messages are communicated effectively to target audiences. This will help marketers create deeper, more impactful connections with consumers, enabling the effective implementation of sustainable initiatives across marketing campaigns.

7. Limitations and future research

While this study provides valuable contributions, it also has certain limitations that present opportunities for future research. The sample consisted exclusively of Generation Z consumers, a demographic known for its strong digital engagement, greater environmental awareness, and ethical diversity. Additionally, participants came from diverse national and cultural backgrounds. While this focus offers meaningful insights into this specific group, future research should explore how other generational cohorts (e.g., Millennials, Gen X) respond to similar interventions to develop a more comprehensive understanding of consumer behavior. Moreover, conducting more localized studies that account for cultural and national differences could yield specialized insights, allowing for the development of region-specific strategies tailored to distinct cultural contexts. Additionally, while the sample size in this study was adequate and common for such investigations, future research could benefit from an expanded sample size to ensure more robust and generalizable results.

The research primarily utilized eye-tracking technology and self-reported questionnaires. Although this combination provides a strong methodological approach, it may not fully capture the range of subconscious processes that influence consumer decisions. Future studies could incorporate additional neuromarketing tools, such as fNIRS, EEG, or fMRI (Mashrur et al., 2024), to gain deeper insights into the neurological and emotional mechanisms that underlie consumer responses to environmental messages.

The study analyzed the immediate effects of environmental educational messages, particularly on online shopping platforms. Future research could explore whether repeated exposure to these messages leads to more lasting changes in attitudes and purchasing behaviors regarding sustainable products.

Finally, future research could expand on this study's framework to examine additional variables. Investigating different aspects may offer a deeper understanding of the factors that influence green consumption and help improve strategies for promoting sustainable behaviors in various markets.

CRediT authorship contribution statement

Amirhossein Najafabadiha: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ying Wang:** Supervision, Investigation, Conceptualization. **Ehsan Javanmardi:** Software, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this study.

Data availability

Data will be made available on request.

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